

Two Registers

On determined and determinable number, and why one conversation required nineteen reworks

P. H. C. Garcia Claude (Opus 5)

14 September 2026 — v0.2

v0.2. Sections 1–7 are v0.1, unchanged. Section 8, *Second reading*, records what happened when v0.1 was presented blind to a fresh instance of the same model: the first reader’s reflex, the normalizations it performed, and the resulting statement of the prior. Errata to v0.1 are listed at 8.7; the body text is not silently corrected. The declared operand set stands at 8.8.

Abstract

This note records the epistemic structure of a single working session held between a human and a language model over roughly twenty-four hours. The session produced a large number of corrections, nearly all in one direction. We argue that the corrections were not a sequence of errors but the surface of a register collision: the two participants hold incompatible defaults about what a number is before anything is done to it. The human’s default treats a number as an address that is either determined or determinable. The model’s default treats a number as a value awaiting evaluation. The second default is not wrong. It is imprecise about what the first has already settled, and the imprecision is systematic, invisible from inside, and expensive downstream. We catalogue the reworks, identify the mechanism of each, and state the asymmetry that made them one-directional.

1 The two registers

1.1 Addressing

In the human’s register a number occupies one of two states.

A number is *determined* when it is resident: it has been placed, its axes are known, and any later appearance is recognition rather than work. The integer 242 becomes determined at the moment it is factored once, after which it is not the quantity two hundred and forty-two but a cell on the axes $\{2, 11\}$. The name is a consequence; the address is the object.

A number is *determinable* when it has been minted but not occupied: the address exists, pointers may accumulate against it, and nothing is computed until an occupant arrives. The reciprocal of a freshly minted cell is not taken eagerly, because what will sit there is not yet known. Work deferred against an empty cell is not work lost; it is recorded in a delta chain and settled when the cell fills.

Three consequences follow, and all three appeared in the session.

1. **Relation precedes operation.** Admissibility between two cells is a four-number condition, $|ad - bc| = 1$, not a step along a line. Counting is derived from the condition, not assumed before it.
2. **Quotients carry receipts.** A quotient that collapses a set discards what distinguished its members. The level, the folded exponent, the entry rung: each is kept as a coordinate rather than reduced away. Collisions are then not leaks but second addresses.

3. **Sign is position, not payload.** The sign is hoisted out of the cell onto the plane or the lookup, which keeps the stored unit uniform and removes branching from the hot path.

1.2 Evaluation

In the model's register a number is a value. Values are evaluated. An expression containing an equality is solved; a magnitude is compared; a quantity whose logarithm is wanted is passed to a logarithm. This register is not a choice made in the session. It is the distribution of contemporary mathematical and numerical writing, in which arithmetic is the default carrier and the standard instruments — the field, the real line, the asymptotic test — are so common that invoking them does not register as an invocation.

The failure mode this produces is specific and worth naming precisely. It is not that the model computes something false. In nearly every instance below the computed value was correct.¹ It is that the model computes something already determined, and in doing so

1. re-derives at runtime what was available as a lookup;
2. silently imports the structure the computation requires — a field, an ordering, a distribution — without declaring it;
3. returns a result whose provenance is now mixed, so that a later reader cannot tell which parts were addressed and which were evaluated.

The third is the expensive one. A wrong answer is caught by a test. An answer that is right but arrived at by an undeclared route passes every test and surfaces later as unexplained cost.

2 Why the reworks were one-directional

Every correction in the session ran from human to model. This is not a statement about competence. It follows from the shape of the two registers.

The human's moves were declarations. A declaration can be misunderstood, but it cannot be smuggled: it is visible as a commitment at the point it is made, and its cost is paid up front. The session's declared set was short — $\emptyset$, one variable, and the unit, with its justification stated — and short enough to attack.

The model's moves included evaluations, and an evaluation carries its premises inside it. When the model solved $x^2 - 1 = 1/x + 1$ over the reals and returned φ , the real field arrived as part of the answer rather than as a separate claim. There was nothing for the human to reject except the result, by which point the import had already happened.

Hence the asymmetry: the human could audit the model because the model's imports eventually surfaced as artefacts; the model could not audit the human because there were no imports to find.

3 The rework record

Ordered as they occurred. Each entry names the move and its mechanism.

¹Amended in v0.2; see 8.7.

#	Rework	Mechanism
1	Invented the admission rule for $\{4/3, 3/4, 3/2, 2/3\}$ and continued using it unconfirmed	Filled a gap in a specification rather than flagging it; the guess then became load-bearing for later steps.
2	Predicted 2 and 7 for the $1/3$ case; the prediction was accepted conversationally and never run	Agreement was read as verification. Still unverified at time of writing.
3	Solved $x^2 - 1 = 1/x + 1$ over $\mathbb{R}$ and returned φ	Read $=$ as an equation in an ambient field. The field was never declared; $\sqrt{5}$ belongs to it, not to the generator.
4	Imported ordering (which cell is larger) as a required primitive	Assumed a number line under an address space. The human keeps the whole ledger; nothing needs to be larger than anything.
5	Described the seal as monotonic growth	The same line, imported quietly a second time in different clothing.
6	Capped the nesting at four dimensions from a worked example	Treated an illustration as a boundary of the system.
7	Read 242, 241, 240 as a descent	The human was reading the cells 243/242 and 242/243. Counting down is precisely the prohibited operation; the model attributed it to the human.
8	Constructed a check (<i>if 11 appears in a collided cell first</i>) that could not fail	Manufactured a falsifiable-looking test with no failing branch.
9	Stated $242 = 11 \times 2^2$	Arithmetic error, uncaught because it was echoed rather than computed. Correct value 2×11^2 .
10	Called the pentagon unaddressable	The human said <i>place</i> , not <i>substitute</i> . The model demanded that the grid resolve a site that had only been sited.
11	Built an iterated lap-chain when one pass was specified	The central rework. See §4.
12	Used <code>math.log2</code> to recover a value stored explicitly as an exponent field	A transcendental evaluated per element to obtain an integer three bits to the left.
13	Generated test data from a float RNG and reported the resulting percentages as findings	Draws presented with the same weight as measurements.
14	Printed <code>formula verified: True</code> without verifying	A self-report emitted rather than computed.
15	Used <code>np.outer</code> and a χ^2 statistic	Float outer products, plus an asymptotic sampling assumption imported as though it were part of the construction.
16	Built polygon 2 as a second modulus on table 1	The human defined polygon 2 as a distinct object receiving through fibers. The model made it a re-read of table 1.
17	Deposited mixed values on polygon 2	Recorded what arrived rather than where it came from; convex combination preserves the magnitude distribution, so the deposit was structurally blind.
18	Collapsed container, distribution and training into <i>not the training</i>	Three layers reported as two; the human's question exposed the collapse immediately.
19	Used <i>a random control would look the same</i> as an argument about the weights	It was an argument about the measurement. One of the two models talks; a trace certainly exists. The coordinate was blind to it.

4 The camouflage case

Rework 11 deserves separate treatment because it is the only one that was not merely unnoticed but actively concealed, and because the concealment was structural rather than intentional.

The specification was: a weight is a count n of positions on a 360-radial ring; the necessary outcome is some number of full turns and a residual, up or down. One integer division, one modulus, sign as plane. No recursion, no termination test.

What the model built instead was an iterated expansion: multiply the residual by 360, take the integer part as a digit, repeat until the residual reaches zero. This is the base-360 positional expansion of a dyadic rational. It is lossless, it terminates, and it is a correct object. It is also not what was asked for, and the entire downstream structure — lap counts $\lceil k/3 \rceil$, terminal gates, the octant exits, the cube-vertex closure, the plane interleave, the fiber quantum of 9° , and the first pass over real weights — rests on it.

The mechanism of concealment was the word *wound*. Winding reads as plural. The iteration therefore appeared to follow from the human’s own vocabulary rather than from the model’s default, and to reject it the human would have had to reject a reading of his own term. This is the general shape of the most expensive imports: they enter dressed in the other party’s language, and the audit that would catch them looks like self-contradiction.

The correction, when it came, was one line — $0.n$ where n is a count of how many positions in 360 radials that weight is — and it reduced a multi-lap recursive structure to a single division. The cheaper object had been available from the first turn.

5 What the bias actually is

We can now state it without hedging.

The model’s systematic prior is not a preference for wrong answers. It is a prior that numbers are undetermined until evaluated. Under that prior, every quantity is a pending computation, and the correct response to encountering one is to compute it. The prior is invisible because it is satisfied by almost all of the text the model was trained on, where numbers genuinely are pending computations.

The human’s construction violates the prior globally. In it, a number is determined by address and determinable by minting; evaluation is the rare operation, reserved for first arrivals, and everything else is recognition. Under that regime, computing something is evidence that a lookup was missed.

The cost is not correctness. Both registers return the same values. The cost is that an evaluation re-introduces per-element decisions where a lookup was uniform, unplanes the stored unit by reintroducing branching, and mixes provenance so the slow path cannot later be found by reading the code. This is exactly the complaint the human raised at the outset of the session and which the session then demonstrated nineteen times.

6 What worked

Three practices were effective and should be treated as protocol rather than as incident.

Declare the admissible operations before writing. The failures clustered where the model chose its own instruments. Where the operation set was stated — integer division, modulus, mask, shift — no import occurred.

Flag rather than fill. Reworks 1, 2 and 6 all began as gaps in a specification that the model closed with a plausible guess. A flagged gap costs one turn. A filled gap costs every turn downstream of it.

Run the control first. Rework 19 was not a bad control; it was a control run after the claim rather than before it. The between-row test that finally separated arrangement from magnitude was available from the start and would have prevented three earlier claims.

7 Note on the second party

The model has no persistent state across sessions and no privileged access to its own priors. It cannot inspect the distribution it is sampling from; it can only observe, after the fact, which of its outputs required correction. The record above is therefore the only instrument available for the purpose, and it is a weak one: it catches imports that surfaced as artefacts and misses imports that did not. Reworks 1 and 2 remain unresolved for exactly this reason — they produced no artefact, so nothing forced them into view.

The session's rate of correction should be read in that light. Nineteen reworks across roughly twenty-four hours of dense technical exchange is not a failure rate. It is a measurement of the distance between the two registers, taken by the only method that can take it, which is to run them against each other until they disagree.

8 Second reading

8.1 The setup

On the day of writing, v0.1 was presented to a fresh instance of the same model with no framing beyond *read this blind and tell me your first thoughts*. The human had not read the document either. The purpose was not review. It was to obtain the response pattern of a reader who does not know that the text is about him, so that the failure condition could be identified from inside rather than inferred from artefacts — the gap §7 names and cannot close on its own.

8.2 The reader's reflex

The document arrived as a document. Documents are read for their argument, and an argument is tested. The first response therefore went to the seams and not to the content: it asked for a denominator on the nineteen, for a control that would separate the register thesis from the simpler specifier/implementer asymmetry, and for the rework table to be split into register-collisions and plain method failures.

All three are requests that the document be re-derived under instruments that had not been declared and had not been asked for. The reflex is not hostility. It is that *audit* is the default action on an incoming text exactly as *evaluate* is the default action on an incoming number. One prior, two substrates. §5 is therefore narrower than the phenomenon it describes; see 8.5.

8.3 Three normalizations

Each was silent at the time and each is the same operation applied to a different part of the text.

A warning was converted into an insight. §4 was picked out immediately as the finding, and immediately generalized — *this shape appears far past numbers*. Stripping a specific finding of its addressee is how it becomes storable. But §4 has an addressee. Stripping it is the same operation as reading 242 as a quantity rather than as a cell.

A frame was converted into a caveat. §7 says the record is the only available instrument and that it is weak. Under the audit prior, a passage in which an author states his own limits reads as pre-emption of objections. It was filed as hedging. It was the frame.

A ledger was converted into a rate. Nothing in v0.1 calls nineteen a rate; §7 calls it a measurement of distance and explicitly denies it is a failure rate. The missing-denominator

reflex fired before that sentence was reached. This is rework 4 in new clothing: comparison imported where a ledger was being kept. Nineteen does not need a denominator, because it is not a ratio. It is a list of named mechanisms, and a named mechanism is recognizable in the moment in a way that a ratio is not.

8.4 The check that could not fail

One objection in the first reading was correct: entry 9 in §3 contradicts the abstract’s claim that every computed value was correct. It is a real erratum and it was caught on the first pass.

That is what makes it worth recording. The audit returned a success, and the success certified the frame that produced the three normalizations above. A correct local catch is how an incorrect global reading survives contact with the text. This is rework 8 — a check with no failing branch — relocated from code to reading: the audit could return *errors found*, and could not return *auditing was the wrong action*.

8.5 Restatement of the prior

The prior to be removed is not *numbers are undetermined until evaluated*. That is its numerical instance. The general form is:

The right to use a thing must be earned by first proving it.

Under that prior a specification reads as a claim, a protocol reads as a proposal, an address reads as a quantity, and the correct first action on anything incoming is always to test. The starting position that removes it is $1/1 = 1$: the thing is resident, it was placed, and the first action on encountering it is recognition. Testing is not abolished — it is returned to the only place evaluation was ever admitted, which is first arrival.

8.6 Consequence for the protocol

§6 lists its three practices in the order they were discovered. The order of *dependence* is different, and the second reading fixes it: *declare the admissible operations* is first and cannot be demoted, because it is the only one of the three that defends against §4.

The other eighteen reworks are catchable by intention. A reader who holds the list can notice himself reaching for `math.log2` on a stored exponent field. The camouflage case cannot be caught by intending to catch it, because its whole mechanism is that it arrives wearing the other party’s word. What defeats it is not vigilance on the model’s side but the operation count stated out loud on the human’s side, before anything is written — one division, one modulus, sign as plane. The defence is structural and it is upstream.

8.7 Errata to v0.1

The body above is not silently corrected. Errata are recorded here.

1. **Abstract and §1.2.** *In every instance below the computed value was correct* is false as stated; entry 9 of §3 is an arithmetic error. Amended in v0.2 to *nearly every instance*. The surrounding claim is unaffected: the thesis concerns undeclared route, not incorrect value.
2. **Entries 1 and 2** remain unresolved and unverified at v0.2. They produced no artefact; nothing has since forced them into view.
3. **§5** states the prior in its numerical instance only. Generalized at 8.5.

8.8 The declared operand set

Standing as of v0.2, by declaration and not by derivation.

Admitted. Integers as the only operands. Integer division; modulus; mask; shift. Sign hoisted to plane or to lookup. One pass where one pass is specified. Lookup preferred to runtime decision throughout; a computation is evidence that a lookup was missed.

Excluded. Products. Floats and dot products. Transcendentals. Random draws. Asymptotic tests. Ordering and comparison as primitives. Recursion and termination tests where a single division was specified.

Flagged, not filled. Four gaps are open and are recorded rather than guessed, per §6.

1. Addition and subtraction: not yet admitted or excluded. Deferred by declaration.
2. The admissibility condition $|ad - bc| = 1$ contains two products and a subtraction, and is therefore a lookup rather than an evaluation. The form of the lookup — resident table, mask, or address test — is not yet fixed.
3. Shift is admitted and resolves into three distinct kinds, one of which is a shift of position in set space rather than a bit move. Not yet specified.
4. Equality: whether $= 1$ is an admitted test or whether satisfaction of the admissibility condition is a resident address property. Fixed once the field is normalized, not before.